Frontier



[Heriot-Watt University]

R&D application

2026

General Application

The General Application applies to everyone; all applicants should complete this section

Public section

The content in this section (answers to questions 1(a) - (e)) will be made public on the Frontier GitHub repository after we announce funding decisions in the summer and fall. Include as much detail as possible but omit sensitive and proprietary information.

Organization name

Research Centre for Carbon Solutions, Heriot-Watt University

Organization location (we welcome applicants from anywhere in the world)

Edinburgh, United Kingdom

Name(s) of primary point(s) of contact for this application

Prof. Phil Renforth and Dr. James S. Campbell

Brief organization description, including organization type (e.g., for-profit, NGO, research institution/university) <20 words

The Research Centre for Carbon Solutions is an interdisciplinary world-leading engineering group, delivering technological innovation needed to meet carbon targets.

1. Public summary of proposed project[1] to Frontier

a. **Project title** <20 words

Alkali Carbonate Technologies for CO_2 Removal (ACTOR)

[1] We use “project” throughout this template, but the term is not intended to denote a single facility or experimental setup. The “project” being proposed to Frontier could include multiple facilities/locations for research activities.

b. **Project summary and field impact:** Provide an overview of your core research goal and the benefits you think it could provide to the broader CDR field. Tell us how your project meaningfully advances knowledge on at least one of the priority innovation areas identified in the RFP. What needs to happen for your project to unlock step-change advancements in cost and scale for this innovation area? Please include figures and system schematics, and be specific but concise. <1000 words

Naturally occurring sodium (Na)- and potassium (K)-rich carbonate minerals have previously been overlooked as a feedstock for geochemical based CDR. This is because a) they were thought to be insufficiently abundant to make a meaningful climate impact at scale, and b) current production costs are prohibitively high. As such, CDR pathways have focused on Ca- and Mg-rich feedstocks such as limestone and silicates.

Our initial work has shown that Na carbonates are substantially more abundant than previously thought and could provide sufficient feedstock for a 0.5-1 $GtCO_2$/yr CDR pathway for 100 years. Furthermore, contemporary production methods are designed to meet market high purity requirements, with opportunities for significant cost reduction (see Campbell et al., accepted).

Our core research goal is to determine whether naturally occurring sodium- and potassium-rich minerals can unlock lower-cost, faster, and more scalable CDR than today's dominant calcium- and magnesium-based approaches. Our hypothesis is that naturally occurring Na- bicarbonates and carbonates such as trona, could open a new frontier for gCDR through two pathways (Figure 1). First, ocean alkalinity enhancement (OAE), in which Na carbonates are attractive because they are far more soluble than limestone, Mg-oxides, and silicates, but are less likely to cause pH spikes associated with strong bases (NaOH, KOH, $Ca(OH)_2$), thus reducing secondary precipitation. Secondly, Na or K carbonates could be a feedstock for ambient (mineralisation) oxide-looping direct air capture (DAC), in which they can be thermally or electrochemically upgraded at substantially lower energy requirements than the calcination required for $CaCO_3$-to-lime conversion.

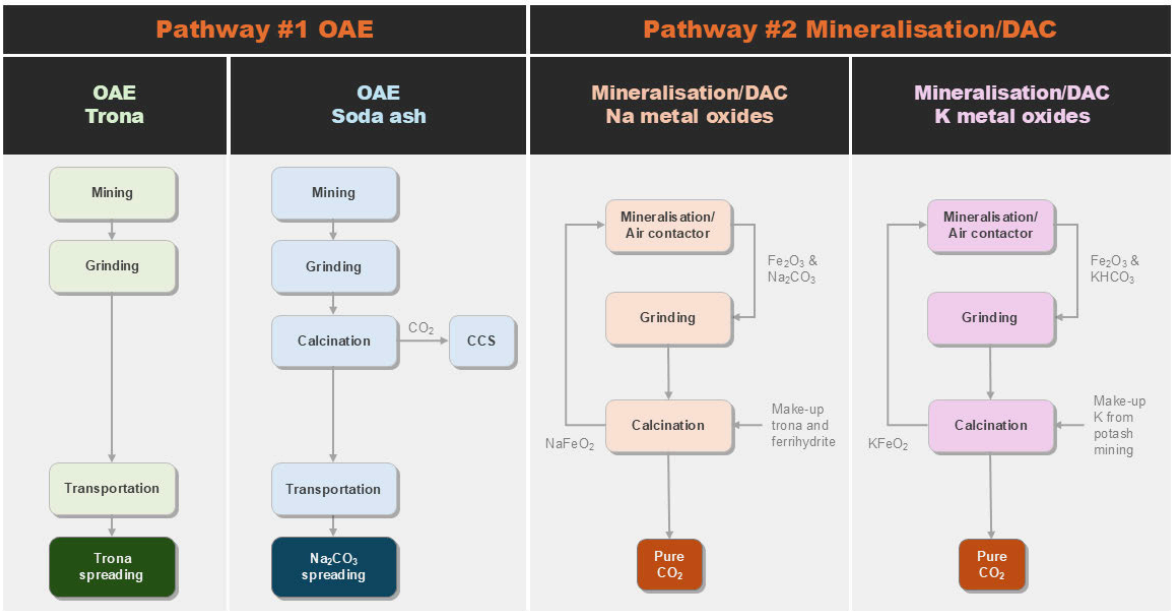


Figure 1: A schematic of the 2 pathways (OAE, and oxide-looping mineralisation DAC, and their

variants) that will be explored in this project.

This work directly aligns with three Frontier target innovation areas: “Strategies to maximize alkalinity addition rates for OAE”, “Enhancing surficial mineralization”, and “Novel CDR approaches with breakthrough potential”.

There are four unknowns that are required to unlock Na/K carbonates for gCDR:

1. **Mapping/resource assessment:** we previously identified over 100 Na carbonate deposits globally, but only 16 of these had quantified resource estimates (equating to 200 Gt of minerals). It is not possible to assess a feasible pathway to scalability without fully mapping and characterising this resource. We will expand our current inventory of sodium bicarbonates/carbonates to higher spatial and compositional resolution, including impurities and likely extraction constraints.
2. **Methods of extraction, processing and upgrading:** Currently, Na carbonate is extracted from deposits using the conventional (resource inefficient) room-and-pillar method. Increasingly, solution mining (injection and recovery of hot alkaline water) is used to maximise resource extraction. The recovered material is processed through crystallisation and thermal treatment (steps which are responsible for > 95% of the overall cost) to produce high purity Na carbonate suitable for the market. Optimising production for geochemical CDR offers opportunities for lowering cost, but potentially increasing the impurities (primarily NaCl and Na_2SO_4) in the final product.
3. **Reaction in seawater (for OAE):** Pure Na carbonate has been extensively used for OAE research, given that it is easier to handle in the laboratory and dissolves quickly in seawater. However, relaxing the purity requirements could provide a means to significantly reduce the cost. We will therefore combine carbonate-system modelling with controlled seawater experiments on material acquired from project partners to determine realistic CO_2 uptake efficiency, precipitation thresholds, and sensitivity to temperature, dosing regime, and receiving-water chemistry.
4. **Reaction with atmospheric CO_2 (for DAC):** During our preliminary study, we identified highly reactive Na/K/Fe-containing phases that may offer faster air capture, good performance in low-humidity conditions, and lower-temperature regeneration than lime-based looping systems. The key uncertainty is not initial reactivity, but cyclic stability: repeated carbonation-regeneration performance, make-up requirements, sorbent degradation, and the availability of low-carbon heat or electrochemical regeneration routes. Here we will identify if artificial Na/K-mineral sorbents offer a genuine step-change rather than an incremental improvement over $CaO/Ca(OH)_2$ systems.

Our work will be directly linked to prospective lifecycle and technoeconomic analysis, with particular attention to transport distance, mine-to-coast logistics, processing intensity, and methane emissions during extraction. If successful, the project would advance the broader CDR field in three ways: by opening a new feedstock base for geochemical CDR, by generating practical evidence on how to maximise retained alkalinity in real waters, and by helping the field avoid dead ends by showing where Na and K pathways truly outperform incumbents and where they do not. That kind of comparative clarity is what is needed to move from promising chemistry to scalable and durable CDR.

c. **Project objectives:** Describe your project in more detail: what are your key research questions and proposed approach for answering them, including methods, project location, and scale where applicable? Tell us why your approach is best-in-class, and how you're differentiated from any other organization working on a similar approach. Please include figures and system schematics, and be specific but concise. 1000-1500 words

Our project asks a simple but high-leverage question: can naturally occurring sodium- and potassium-rich minerals unlock faster, lower-cost, and more scalable durable carbon dioxide removal than today's dominant calcium- and magnesium-based pathways? Most geochemical CDR efforts have focused on Ca- and Mg-rich feedstocks such as limestone, quicklime, brucite, olivine, and silicates. In contrast, Na- and K-rich carbonate minerals remain comparatively underexplored, despite evidence from our preliminary work that some may offer major advantages in solubility, reactivity, lower-temperature upgrading, and potentially novel direct air capture chemistry.

The project is designed as a comparative de-risking program rather than a single-pathway optimisation effort. Our objective is not to assume sodium or potassium materials will outperform incumbent feedstocks, but to determine where they do, why they do, and under what constraints. We will compare naturally occurring sodium bicarbonates/carbonates (e.g. trona) across the full chain of CDR performance: resource quality, impurities, extraction constraints, transport, process energy, methane leakage risk, reaction kinetics, carbonate chemistry, cyclic stability, MRV implications, and cost.

The project is structured around four research questions:

First, what Na/K-rich resources are actually available, in what quantities, and with what compositional and logistical constraints?

Our first work package is a resource mapping and feedstock screening effort. We will upgrade our current inventory of sodium bicarbonate/carbonate resources to higher spatial and compositional resolution, including mineralogy, impurity profiles, extraction mode, distance to deployment locations, and any evidence of associated methane emissions from mining. We will extend this mapping to sodium-rich industrial residues and by-products, especially materials that are abundant but under characterised. Here we will go beyond an estimate of theoretical abundance, but to identify practical feedstocks: materials that exist at meaningful scale, can be accessed with acceptable environmental burdens, and possess chemistries compatible with OAE, IWAE, mineralisation, or DAC applications. We will also map potassium-rich minerals, including potash-bearing evaporites and related K-rich deposits, to assess whether they represent practical feedstocks or upgrading precursors for OAE, IWAE, mineralisation, or DAC applications.

Second, which of these resources are promising once full system constraints are included?

Resources need to be extracted, processed, and transported to be suitable for gCDR. Here we will explore two technology archetypes: a once-through OAE system in which the end of the Na-carbonate supply chain is the distribution of minerals into the ocean, and an ambient oxide looping system for Na and K carbonates. Many sodium carbonate deposits occur in arid inland regions far from coastlines, and transport can dominate cost and emissions. Furthermore, their reaction with CO_2, either through dissolution in seawater or ambient carbonation, remains poorly constrained (see below).

This project will use a comprehensive framework for prospective technology assessment that we

developed previously (Lee Pereira et al., 2025) to include transport distance, mine-to-coast logistics, water requirements, energy source, waste handling, methane emissions during extraction where relevant, and sensitivity to different deployment contexts. The output will be a ranked, decision-oriented view of which feedstocks and process routes appear plausible (in terms of economic cost and environmental impact) for scaled CDR and which are unlikely to remain competitive.

Conventional methods of Na-carbonate production involve leaching, crystallisation, and thermal purification/upgrading. This production pathway is driven by the need to remove impurities within the extracted materials (primarily NaCl, Na_2SO_4). Permitting these impurities may result in substantial cost savings, but potentially complicate environmental performance.

Thermal Na/K transformation pathways occur at substantially lower temperatures than conventional lime production. For example, selected sodium bicarbonate/carbonate systems may be upgraded without the thermal intensity required to convert $CaCO_3$ to CaO (>900°C), or thermal activation of serpentines (>600°C), for example. That matters because high-temperature processing is a major cost and emissions driver in many mineral-based pathways. However, a lower theoretical calcination temperature does not automatically imply a better system once heat source, kinetics, mass yield, and transport are included. We will therefore evaluate both thermal and electrochemical upgrading routes both experimentally and through system analysis. These upgraded products will be assessed in terms of added reactivity, handling, net carbon benefit, and whether the performance gain is large enough to justify the added process complexity.

Third, how effective are sodium carbonate pathways for OAE and IWAE under real aqueous chemistry?

This is the core scientific question for the project's best fit with Frontier's OAE/IWAE priority area. The theoretical potential CO_2 removal from Na_2CO_3 addition is straightforward to estimate, but real seawater and inland-water systems are governed by carbonate equilibria, mixing dynamics, and secondary precipitation. These factors determine how much added alkalinity remains available for atmospheric CO_2 uptake versus being lost locally as carbonate minerals. We will address this through a combination of modelling and controlled experiments. On the modelling side, we will use carbonate-system tools and equilibrium calculations to map expected behaviour across salinity, temperature, alkalinity background, and dosing conditions. On the experimental side, we will conduct bench-scale seawater and inland-water tests to measure dissolution behaviour, pH excursion, precipitation thresholds, and effective retained alkalinity under different addition strategies. These tests will focus especially on comparing Na-rich feedstocks against more familiar carbonate/alkaline materials in terms of solubility, near-field chemistry, and operational robustness.

This work is important because sodium carbonates may occupy a valuable middle ground: far more soluble than limestone and many silicate feedstocks, but potentially milder in their local pH effects than stronger caustics such as lime or NaOH. If confirmed, that could make them easier to disperse, easier to monitor, and less operationally constrained in open-water or inland-water settings. At the same time, our approach is deliberately conservative: we will quantify where carbonate equilibria reduce effective CO_2 uptake below stoichiometric maxima, and where secondary precipitation, impurities, or logistical limits erase the apparent advantages.

Fourth, do Na/K-rich minerals enable a new of mineral-based oxide looping direct air capture?

In preliminary experiments motivated by evaporite and saline mineral systems, we identified highly reactive Na/K/Fe-containing materials that appear to carbonate in air more rapidly than lime under

some conditions and may also regenerate at lower temperature. We are particularly interested in materials that perform well in low-humidity environments, where conventional slaking-dependent systems are disadvantaged. The central question is not initial performance but durability: can these materials maintain capacity across repeated carbonation-regeneration cycles, and what make-up rates would be required to offset sintering, deactivation, or phase change? We will therefore perform cyclic laboratory tests to measure uptake rate, working capacity, regeneration energy, stability, and sorbent attrition/degradation over repeated cycles. If these materials retain performance, they could represent an important alternative to $CaO/Ca(OH)_2$-based mineral looping systems. If they fail under cycling, that will also be a valuable outcome for the field.

Project management

The project is primarily laboratory- and analysis-based. The mapping, geospatial screening, TEA, and LCA components are global in scope. The experimental work will be conducted at bench scale using representative natural samples, synthetic or collected seawater/inland waters where relevant, and controlled laboratory reactors or exposure systems. This is the right scale for the present stage: the goal is to resolve first-order uncertainties and identify the few pathways that warrant later pilot-scale investment.

Our approach is best-in-class because it integrates disciplines that are usually separated. Today, most OAE teams focus on delivery, monitoring, and field performance of a small number of known alkalinity sources. Most DAC teams focus on engineered sorbents or calcium looping systems. Resource geology groups may map minerals, but usually not with direct linkage to CDR performance or MRV. We connect these layers in a single program: resource mapping, chemistry, process design, environmental constraints, and comparative technology assessment. That allows us to answer a more consequential question than “does this material react?” The question is, “Is this a scalable, defensible CDR pathway once reality is included?”

We are differentiated from other organisations in three ways. First, we are focusing on a feedstock class that remains outside the mainstream CDR portfolio despite plausible large-scale availability and compelling chemistry. Second, we are comparing multiple branches of the sodium/potassium platform rather than advocating for one predetermined solution. Third, we are treating OAE/IWAE and DAC not as disconnected silos but as related opportunities arising from the same family of feedstocks and upgrading methods. That cross-platform perspective increases the chance of identifying unexpected high-value pathways and reduces the risk of overcommitting to a branch that fails once broader system constraints are evaluated.

The expected outputs are practical and decision-relevant: a higher-resolution inventory of Na/K feedstocks and residues; a screened set of candidate materials suitable for different CDR uses; experimentally constrained estimates of retained alkalinity and CO_2 uptake in aqueous systems; performance data for new DAC-relevant sorbents; and comparative TEA/LCA outputs showing when sodium/potassium routes outperform incumbent Ca/Mg options and when they do not. That combination of evidence would allow Frontier and the broader CDR field to make much better decisions about where to focus future pilots, capital, and research effort.

d. **Risks:** What are the biggest risks to your project’s success and how will you mitigate those? Include scientific; technical; project execution; measurement, reporting and verification (MRV); ecosystem; financial; and any other risks. <500 words

While much is known about the reaction behavior of pure Na and K carbonates, the reaction of realistic materials with CO_2 is not well constrained across environmental conditions. This may control the technical performance of these approaches, and as such it is the focus of much of the work in this project. For OAE/IWAE, theoretical CO_2 uptake may be reduced by impurities and secondary mineral precipitation, so removals could be materially lower than simple estimates. We will mitigate this by combining modelling with laboratory experiments designed to quantify realistic operating windows, retained alkalinity, precipitation thresholds, and cyclic sorbent stability before advancing any pathway.

A key technical risk is that upgrading or handling these materials may prove more difficult, energy-intensive, or emissions-intensive than expected. Lower calcination temperatures or electrochemical routes may not deliver sufficient practical advantage once kinetics, process integration, and heat or electricity requirements are included. We will mitigate this by comparing direct-use, thermal, and other transformation processes on a consistent technoeconomic and lifecycle basis, with explicit go/no-go criteria. We will work closely with project partners (including trona producers, and kiln operators) who will bring practical experience with material handling and processing.

A key project execution risk is that this is a broad comparative program spanning resource geology, aqueous chemistry, process analysis, and material testing. To reduce this risk, we are structuring the work as a staged screening effort: first resource characterisation and laboratory testing, then TEA/LCA, then narrowing to the highest-value branches. This allows early elimination of weak candidates and keeps effort focused on the most decision-relevant uncertainties.

The main MRV risk is that apparent alkalinity addition may not translate into durable atmospheric CO_2 removal at the expected efficiency, particularly in open-water systems. There are also uncertainties around how impurities or local precipitation events affect monitoring needs. We will mitigate this by explicitly measuring retained alkalinity and constraining near-field chemistry.

The principal ecosystem risk is that natural evaporites, alkaline residues, or upgraded alkalis may contain impurities or create localised chemical conditions that are environmentally unacceptable. We will therefore prioritize detailed impurity characterisation, screen out problematic materials early, and focus on feedstocks and dosing conditions that minimise extreme pH excursions and secondary precipitation. While ecosystem impact experiments are premature, our group has extensive experience working with ecologists and biogeochemists (through major EU and philanthropic supported projects), and we will look to further investigate the potential environmental impact of promising candidate materials identified from this project.

The main financial risk is that even if the chemistry is promising, supply chain or processing costs could undermine overall economics. We will mitigate this by incorporating logistics, transport, and extraction-related emissions into the analysis from the outset and by identifying only those pathways that remain credible under realistic infrastructure assumptions.

Overall, the central risk is that sodium- and potassium-based pathways may prove less advantageous than they appear initially. Our mitigation strategy is therefore comparative and evidence-driven: fail weak branches quickly and focus resources on the few pathways that can withstand full system-level scrutiny.

e. **Proposal to Frontier:** Please list your primary research deliverables, project timeline, and requested funding below. If you are selected for Frontier funding, this table will form the basis of contract discussions.

Primary research deliverables *(List tangible outputs of the project: e.g., validated datasets, peer-reviewed models, bench-scale prototypes, or field trial results. Should focus on resolving a specific bottleneck or uncertainty identified in Section 1.a)*	1 An open, high-resolution inventory of Na- and K-rich minerals relevant to gCDR. 2 Bench-scale OAE/IWAE and mineralisation/DAC datasets quantifying retained alkalinity, precipitation thresholds, CO_2 uptake efficiency, sorbent cycling stability, and regeneration requirements. 3 A comparative TEA/LCA of direct-use, thermal-upgrading, and other production pathways.
Field impact *(Briefly describe how these deliverables resolve key bottlenecks for a specific CDR pathway or broader CDR field, e.g. efficiency gains, scaling unlocks, or reducing measurement uncertainty.)*	Together, these deliverables resolve three major bottlenecks: identifying credible Na/K feedstocks, determining whether they remain advantaged once full system constraints are included, and generating realistic performance data for OAE/IWAE and mineralisation/DAC. The output is a decision-ready framework for where sodium/potassium pathways can lower cost and improve scale for durable CDR.
Project timeline & key milestones *(At what point should Frontier consider the research project complete? Include major progress markers. Frontier generally prefers projects with clear readouts within ~18 months.)*	Month 6: feedstock inventory development and characterization complete. Month 12: preliminary experimental work complete, coupled with high-level cost assessment. Decisions made on technology focus of the remaining 8

	months. Month 20: Laboratory experiments and analysis complete. Month 24: Full technology assessment (cost and environmental performance complete)
Total grant request (US$) *(The total funding required to execute the research scope described above. Frontier R&D grants typically range from US$250k–$750k.)*	$631K